ARCH 719

DESIGNING FOR A CHANGING CLIMATE

London Group

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1.0.Definition of Climate

Climate change can be simply defined as long-term change in the average temperature of the Earth including changes in weather patterns. The last century has experienced a rapid increase in temperature on the planet. Worldwide temperatures have increased by 1.28 degrees Celsius between 2015 and 2024 compared to earlier century. Since the early years of 1980, temperatures have proved to be warmer than before, and climate change is the sole reason for this. In 2024, there was a record high temperature increase over taking the existing of 1.28 degrees to 1.5 degrees over pre-industrial times (BBC, 2024).

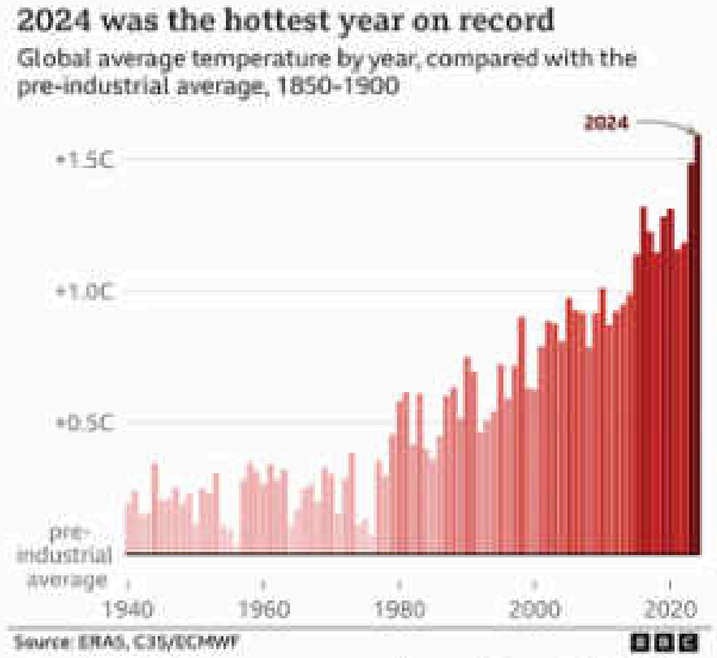


Figure 1. Global average temperature by year (Source: BBC, 2024)

1.1.Causes of Climate Change

- Natural factor

Climate change is influenced by various natural factors, including variations in solar activity, changes in Earth's orbit, volcanic eruptions, and ocean current shifts. For example, increases and decreases in sunspot activity affect the amount of solar radiation received by Earth, while changes in the Earth's orbit and axial tilt, known as the Milankovitch cycle, contribute to long-term climate fluctuations (British Geological survey, 2025) (Morales, 2022).

Apart from the above factors, other critical natural causes such as volcanoes release huge quantities of sulfur dioxide and other aerosols which temporarily reduce global temperatures. El Niño and La Niña are ocean current changes that studies have proven to have effect on precipitation and temperature patterns. All these factors combined have an impact on the Earth's climate. (Schmidt and Carn, 2022).

- Human factor (main causes)

The major cause of climate change is greenhouse gases, and human activities. The known cause of CO_2 is from burning fossil fuels such as coal, oil, and natural gas, as well as deforestation, leading to an increase in global warming. Animal farming, landfills, and rice paddies produce gases called CH_4, while N_2O generally comes from the use of fertilisers and burning of fossil fuels. Also, gases such as hydrofluorocarbons and perfluorocarbons are commonly found in refrigerants and aerosol sprays of different items that are used daily (Filonchyk et al. 2024).

In addition, agricultural and industrial activities change the nature and use of the land and further lead to the emission of greenhouse gases, having negative impacts on the environment. The growth of cities and urbanization is a major cause of urban heat island, affecting microclimates leading to a rise in temperatures, causing climate change (Sovacool et al. 2021).

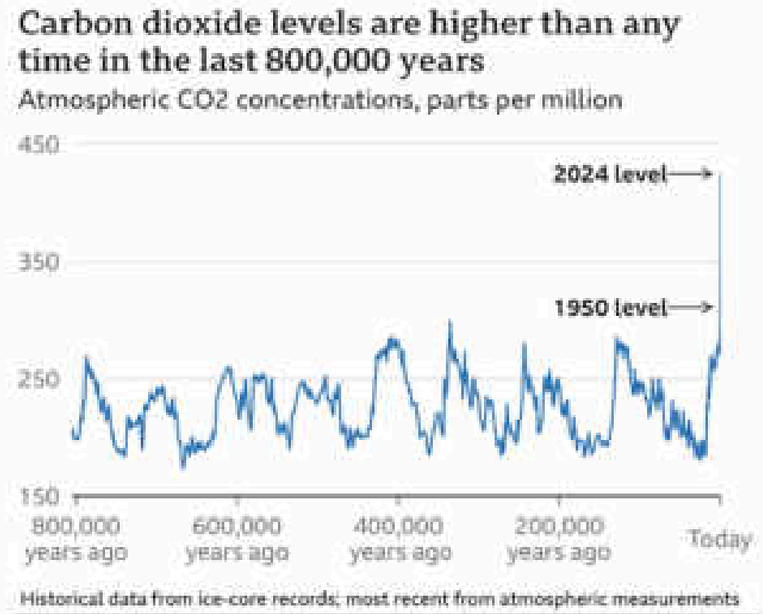


Figure 2. Carbon dioxide levels from 800,000 years ago to today (Source: BBC, 2024)

1.2.An introduction of the causes of anthropogenic climate change

A major part of greenhouse gas emissions can be attributed to anthropogenic causes as several studies have shown, mostly due to population growth and human urbanisation expansion. Recent climate studies indicate that carbon dioxide is extremely high compared to the 278 parts per million that was known to be in the atmosphere before industrialisation. History in climate studies shows that carbon dioxide recordings have exceeded four hundred parts per million in 2016. Verified information from the last 800, 000 years informs that carbon dioxide varies from earlier recordings of 185 parts per million to 300 parts per million in later years. (BBC, 2024).

These human greenhouse gas emissions come from a variety of sources. 78% of all emissions since 1970 have come from industrial operations and the combustion of fossil fuels. The generation of heat and electricity makes up most of the economic industry (25%). Land use, forestry, and agriculture (24%). Sector (21%). Cars and other forms of transportation (14%). buildings (6.4%) and other energy generation (9.6%). Forty percent of greenhouse gas emissions are still in the atmosphere, and more than half of them have happened in the last forty years. In actuality, the biggest rise since 1970 occurred between 2000 and 2010 (BBC, 2024).

Buildings, whether residential or commercial, produce significant volumes of greenhouse gases via activities such as heating, cooling, cooking, electrical appliances, and controlling other building systems. Every day, energy efficiency improves, allowing individuals to accomplish the same or better results while using less energy (IPCC, 2014).

1.3.How computer modelling can be used to predict future changes in the Earth's climate for different regions

Scientists have a means of predicting future climates using a method known as climate modelling and this is based on computer simulations, detailed principles in science and mathematics to predict trends and temperature recordings under different emission scenarios. First, a method must be selected, such as SRES or RCP, which determines the future trends of greenhouse gas emissions. In addition, past meteorological data, terrain, land surface types, ocean currents, and solar radiation are input as initial and boundary conditions. Climate simulations are run with simulation programs for different regions, and they can produce global-scale temperature, precipitation, and wind patterns. Finally, annual average temperature changes and assessments of impacts on buildings and energy can be obtained. Taking the United Kingdom as an example, climate projections of temperature from low, medium and high emissions scenarios' are equivalent to global temperature changes (HCCPR, 2017).

- Explaining the Representative Concentration Pathways (RCP)

The United Nations' Intergovernmental Panel on Climate Change (IPCC) models and predicts climate using standard emission scenarios known as Representative Concentration Pathways (RCPs). They forecast the future patterns in Earth's climate change and show diverse ways that human greenhouse gas emissions may go in the future.

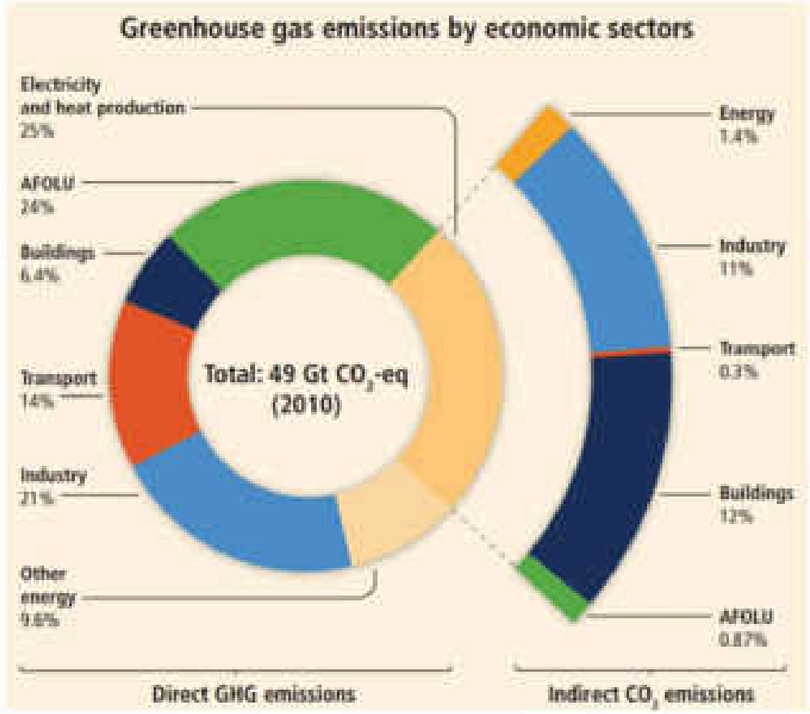


Figure 3. The total anthropogenic greenhouse gas (GHG) emissions of the economic sector in 2010 (Source: IPCC, 2014)

- RCP2.6 (Low Emission Scenario)

If human beings have made successful steps to reduce emissions (e.g., utilising clean energy, large-scale carbon capture technology, etc.), this is the most hopeful course. The amount of greenhouse gases will either decrease or stabilise by the year 2100. The aim is to reduce global warming to less than 2°C to lessen the harm of climate change.

- RCP4.5 (Medium Emission Scenario)

Assuming that some policy measures had been implemented to restrict emissions, around the middle of the twenty-first century, emissions would reach their peak before progressively declining. With the global temperature increasing by about 2.4°C, it falls within the "mild control" method.

- RCP6.0 (Higher Emission Scenario)

Up until the middle of this century, emissions were rising before progressively levelling down. Rather than actively reducing emissions, policy control is comparatively weak and depends on technology advancement. The influence on the climate is substantial, and the extent of global warming is comparatively great.

- RCP8.5 (High Emission Scenario)

If people keep using fossil fuels and don't take any meaningful steps to cut emissions, greenhouse gas concentrations will rise sharply by 2100, and global warming may surpass 4°C. This "catastrophic path" is frequently used to gauge the direst outcomes as seen in Figure 4 (IPCC, 2013).

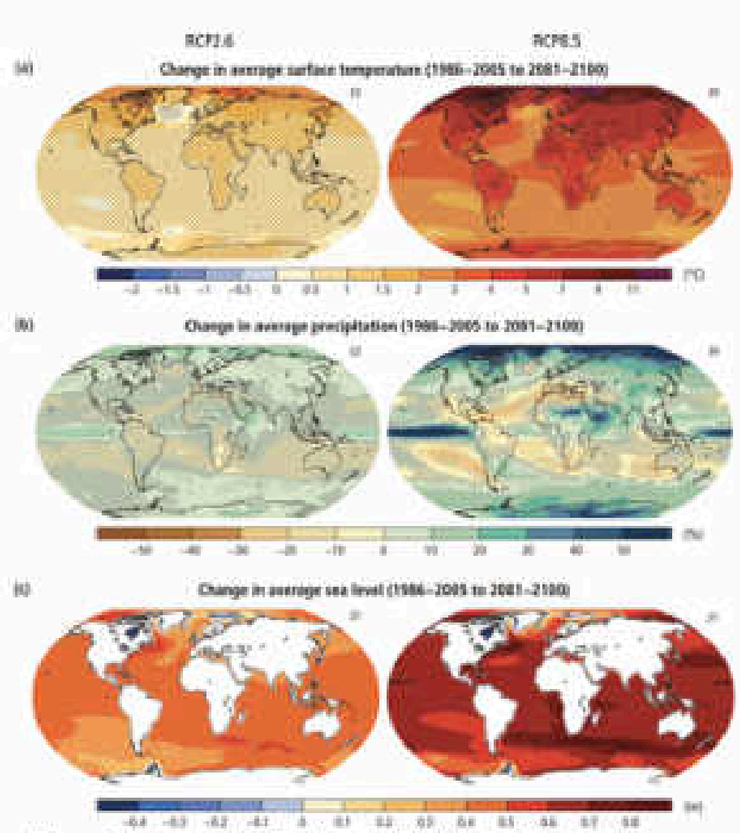


Figure 4.Changes to the Earth under RCP2.6 and 8.5 scenarios. (Source: IPCC, 2014)

1.4 Differences between RCPs and SRES

The Special Report on Emissions Scenarios (SRES) was published by the Intergovernmental Panel on Climate Change (IPCC) in 2000. It describes new scenarios that may emerge in the future and predicts greenhouse gas emissions under these scenarios. The report states that even without explicit climate policies, future emissions will remain highly dependent on human choices: how economic structures are constructed, which energy sources are preferred, and how available land resources are utilised (IPCC, 2000). RCPs use the radiative forcing caused by greenhouse gases, while SRES takes socio-economic development pathways as its core starting point. Second, emissions from specific sources in SRES (such as SO_2 emissions and CO_2 emissions from land-use changes) result in different concentrations, while RCPs simulate climate by directly setting concentrations. SRES has four major scenarios (Figure 6): A1, A2, B1, and B2. A-series scenarios describe a world inclined toward economic development, while B-series scenarios lean toward social and environmental development; scenarios with the 1-series exhibit frequent economic, cultural, and technological exchanges, while the world in the B2 scenarios is less globalised. In 2014, the IPCC adopted RCPs to estimate future greenhouse gas concentrations over the next 100 years or even longer and converted these into increased radiative forcing (IPCC, 2014).

Table 1: SRES and RCPs

Difference	SRES (Special Report on Emissions Scenarios)	RCPs (Representative Concentration Pathways)
Concept	Projecting emissions based on scenarios for future socio-economic development	Set future Greenhouse Gas concentration pathways in the atmosphere and derive possible emission scenarios.
Main Variable	Greenhouse gas emissions (such as CO_2 emissions)	Radiative forcing caused by greenhouse gas concentrations (unit: W/m^2)
Number of Scenarios (Main Category)	Four major categories (A1, A2, B1, B2)	RCP2.6、4.5、6.0、8.5
Assumptions about Human Behaviour	Assuming that future economic, demographic, technological, and other social variables affect emissions	Does not directly describe human social behaviour; concentration is a predetermined result
Current Usage	Belongs to the old system and is gradually being phased out	The scenarios are more comprehensive and include emission scenarios that take climate policy into account.

Figure 5:Comparison table of SRES and RCPs (Source: IPCC, 2013 and IPCC,

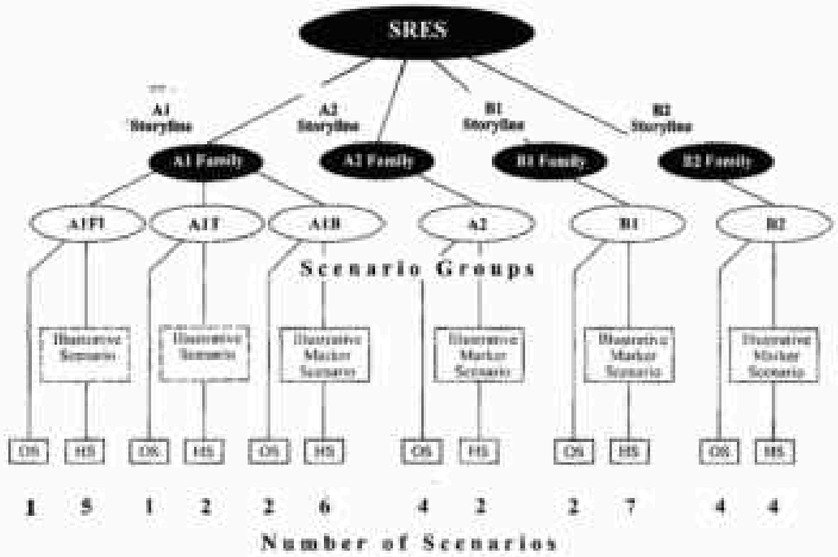


Figure 6:Schematic illustration of SRES scenarios. (Source: IPCC, 2000)

2.0.Project Details

The building that is chosen for this project to study under different future climate scenarios is a commercial building located in London, UK. It is a 5 storey building with first two levels as stores and the rest as offices. The building receives natural light only from the South side. The two zones chosen to be analyzed are located on ground and last floor. The ground floor is chosen due to the benefit of contact diretly with the ground that can perform as a natural insulator and the last floor to ascertain its maximum envelope exposure to the direct sun.

Building Type	Commercial-Office Mixed-Use Building					
Time-table	9am- 7 pm	All day				
Information	**Material**		**Internal heat gain**		**Occupancy**	
	Surface	**Type of Material**	**Source**	**Amount**	**Space**	**Number of People**
	Floor	Reinforced concrete with a concrete slab	Occupants (based on activity)	70-100 W/person	Retail Spaces	15-30 people
	Walls	Concrete panels with limestone finishes	Equipment and lighting	12/20 W/m²	Office area	10-20 people
	glazing	Single-glazed glass -framed in steel	Solar heat gain	5/10 W/m²		
	Roof	Rubber membrane				
	Internal Wall	Plaster Board on Metal Studs				



Figure 7. Image of the chosen building in London

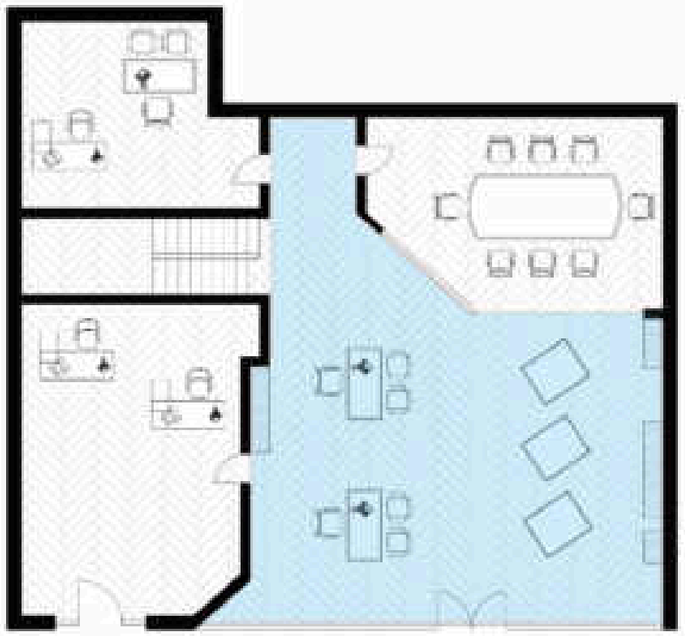
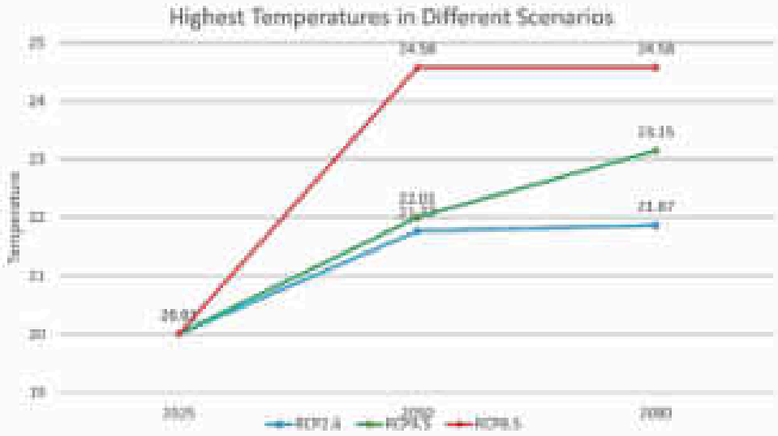


Figure 8.
Store Plan Model (Gf and 1ST Floor)

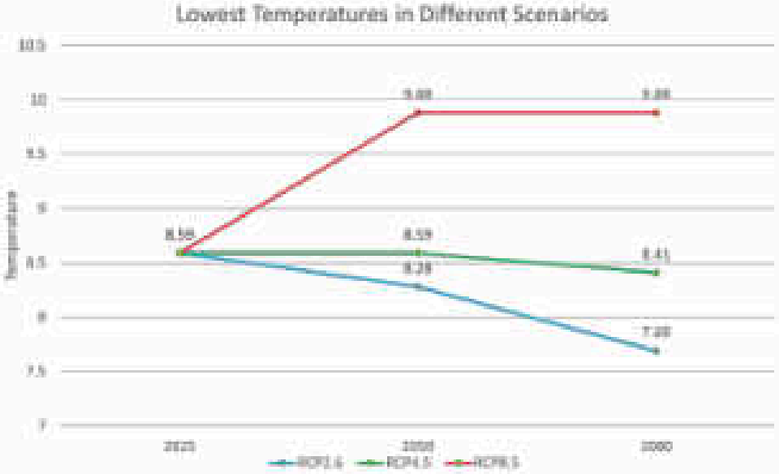


Figure 9.
Office Plan Model (2nd , 3rd and 4th

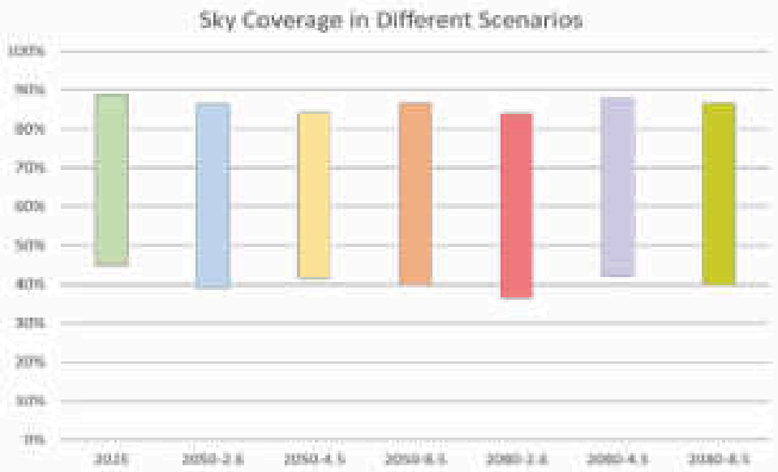
2.1. Climate Scenarios and Representative Concentration Pathways



The temperature at the project's location in London for 2025 at 20.01°C forms the baseline across the scenarios for the highest recorded temperatures. In 2050, the temperature increases under RCP 2.6 at 21.77°C, further increasing to 22.01°C under RCP 4.5, further rises under RCP 8.5 at 24.58°C. The year 2080 has highest record of temperature starting at 21.87°C under RCP 2.6, increases significantly to 24.58°C under RCP 8.5 The data shows a percentage increase of 21.14% between 2025 and 2080 which further agrees with the projections of global warming.



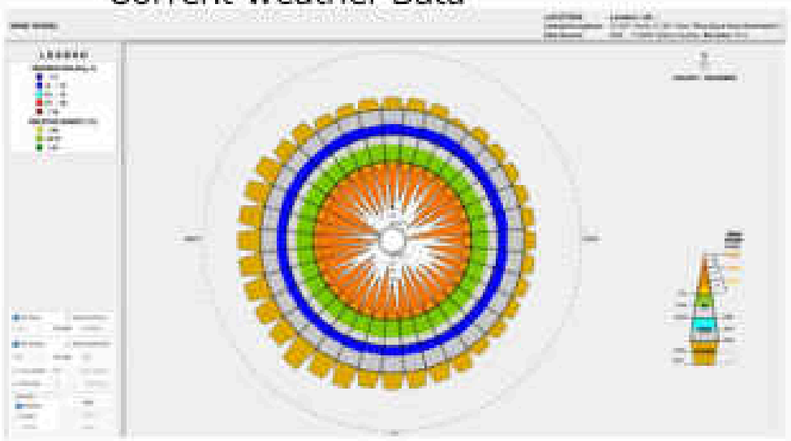
The lowest recorded temperature for 2025 at 8.59°C forms the baseline across the scenarios.In 2050, the lowest temperature reduces under RCP 2.6 at 8.28°C, further increasing to 8.59°C under RCP 4.5, but rises sharply under RCP 8.5 at 9.88°C. The year 2080 has lowest temperature drop of 7.69°C under RCP 2.6, increases slightly under RCP 4.5 at 8.41°C, but it stays very high at 9.88°C under RCP 8.5. A comparative analysis of the data shows a percentage increase of 15.02% between 2025 and 2080 which means minimum temperatures are warming up, which truly aligns with global warming.



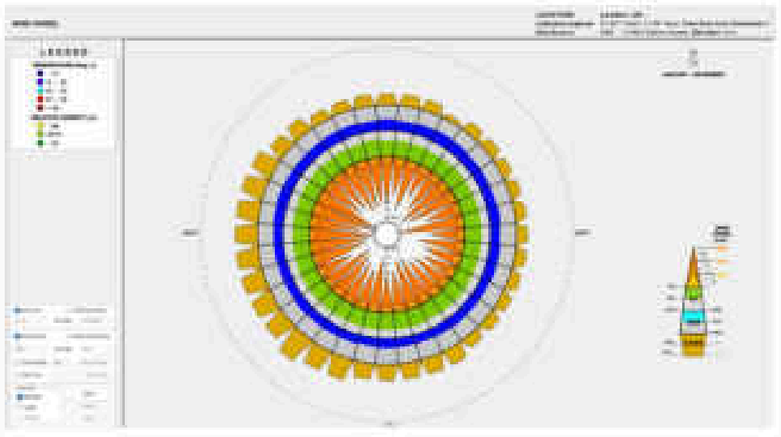
The chart indicates a decline in sky coverage under 2050 RCP 2.6 and RCP 8.5 with values at 2.9 and 1.0. Between 2050 and 2080, sky coverage values fluctuate between 1.3 and 1.0. This informs that by 2080, there would be a more sunlight reaching surface temperatures of the location of the project due to a decline in sky coverage causing warmer temperatures.

2.2 Climate Scenarios - Wind Data

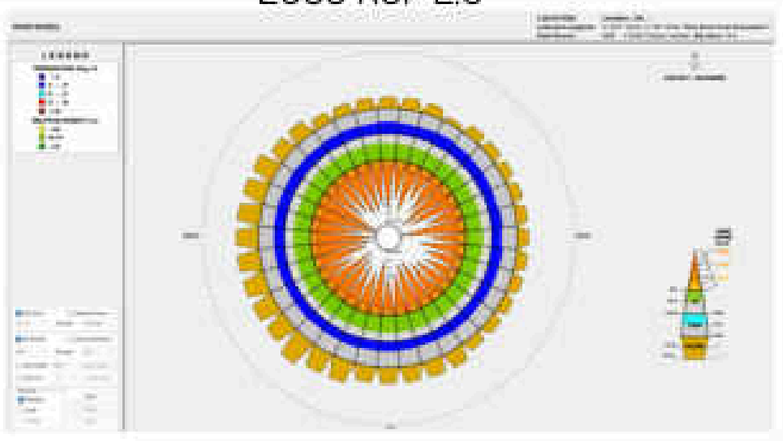
Current Weather Data



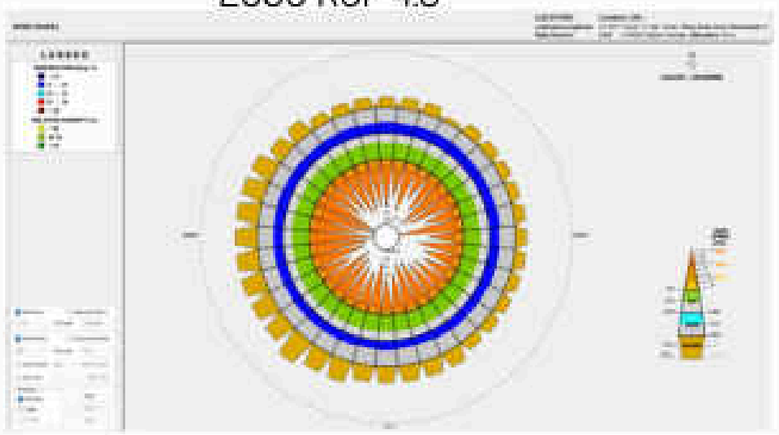
2050 RCP 2.6



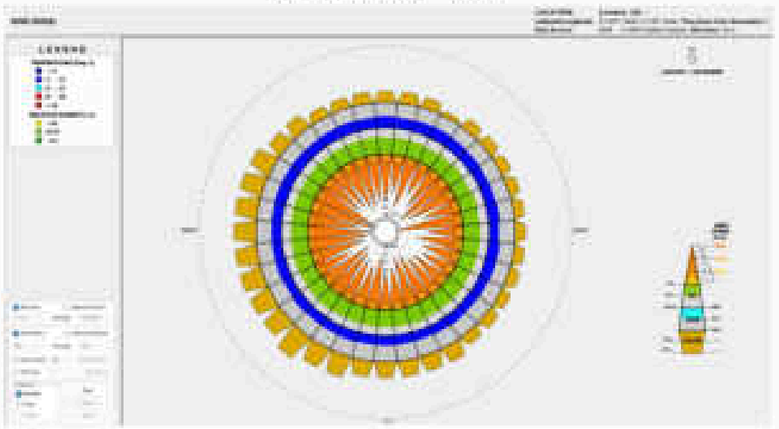
2080 RCP 2.6



2050 RCP 4.5



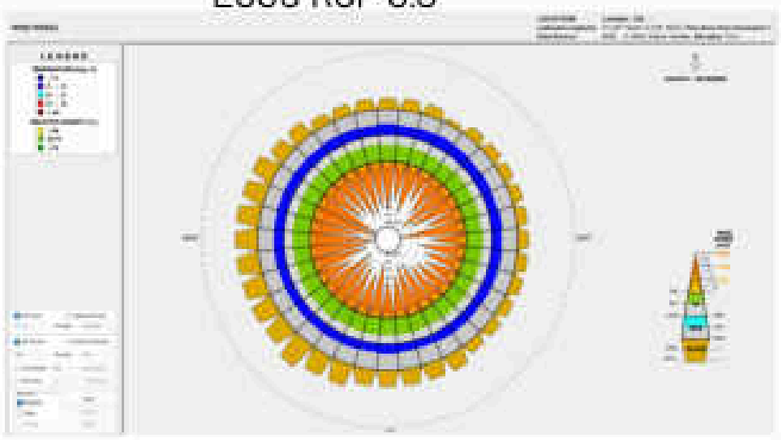
2080 RCP 4.5



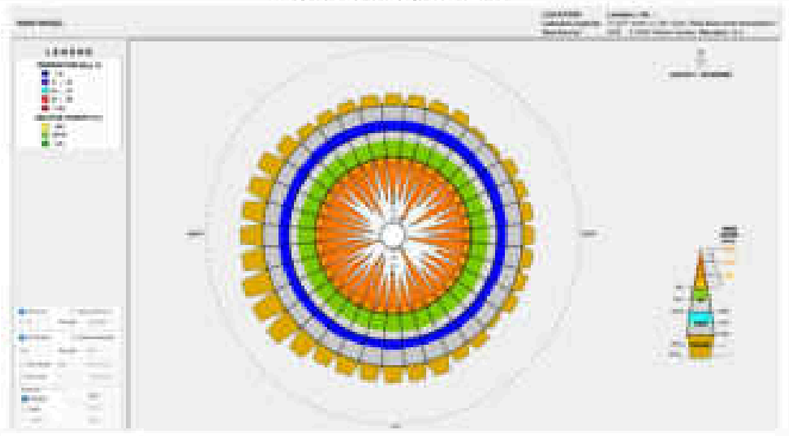
The wind wheels across all scenarios give very similar information regarding direction and speed, indicating prevailing winds majorly from West, Southwest and Northwest at an average wind speed of 3 to 4.5 m/s but peaks at 16m/s.

Other winds from North, South and East have speed between 3 to 3.5m/s, peaking between 14 and 16 m/s. The patterns suggest variable speed where there are more calm days even though there are peaks at 16m/s suggesting high seasonal storms.

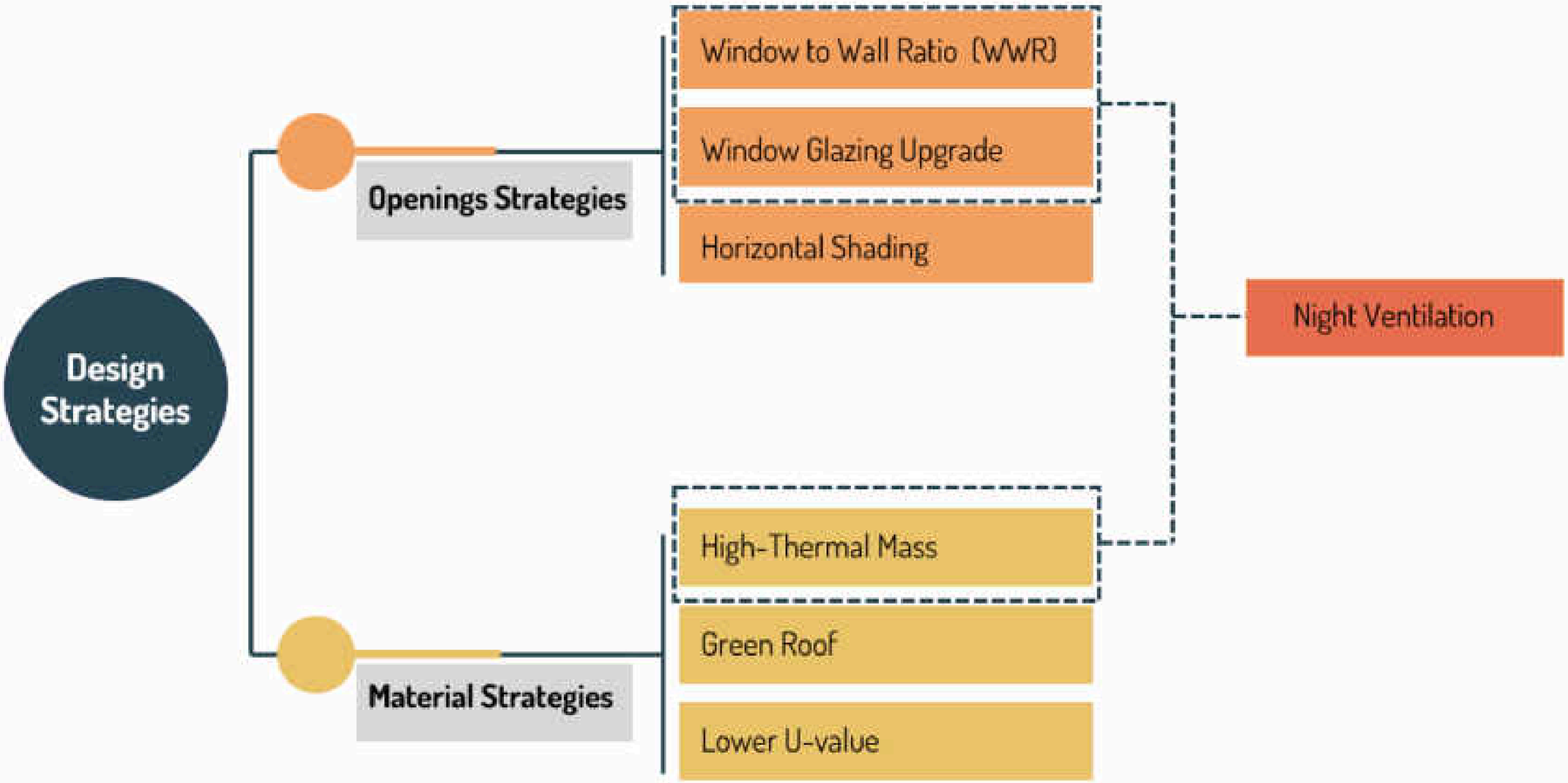
2050 RCP 8.5



2080 RCP 8.5

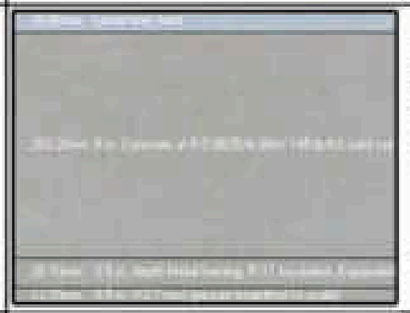
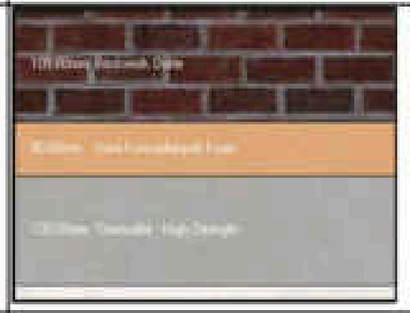
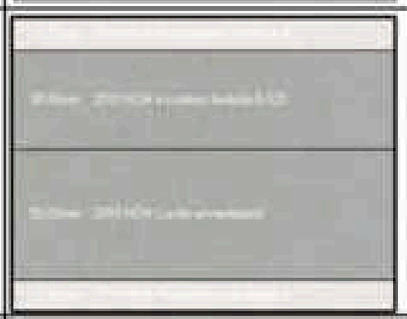
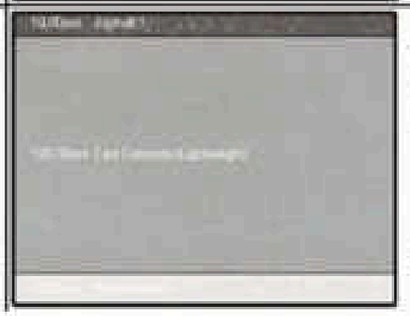
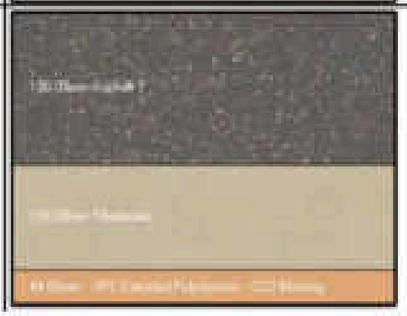
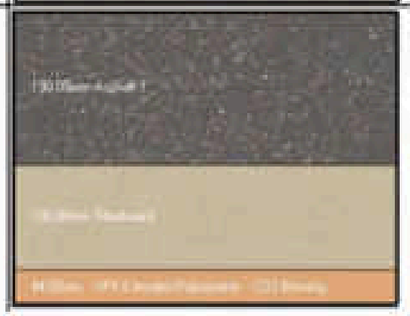


3.0. Summary of Design Strategies



3.1. Material Comparison

Table below displays different materials chosen for the current building with high U-values and excludes insulation. In incorporating the different strategies, materials with high thermal mass and insulation, which had low U-values were selected to improve the indoor thermal environment of the chosen zones. The final selection incorporated external walls with dense insulation and lower U-values.

	CURRENT MATERIALS		HIGH-THERMAL MASS MATERIALS		LOWER U-VALUE WALLS	
FLOOR		100.00 MM Cast Concrete **U-value: 2.929 W/m².K**		100.00 MM Cast Concrete **U-value: 2.929 W/m².K**		100.00 MM Cast Concrete **U-value: 2.929 W/m².K**
EXTERNAL WALLS		20.00 MM Limestone hard 203 MM Concrete 26.10 MM 3.5 in Metal framing 12.70 MM 0.5 gypsum board **U-value: 0.922 W/m².K**		105.00 MM Brickwork Outer 25.00 MM Air gap 10 MM 25.00 MM Phenolic Foam 100.00 MM Concrete Block (Heavy Weight) 13.00 MM Plaster **U-value: 0.825 W/m².K**		105.00 MM Brickwork Outer 50 MM Urea Formaldehyde Foam 100.00 MM Thermalite- High Strength 13.00 MM Plaster **U-value: 0.465 W/m².K**
INTERNAL WALLS		13.00 MM Gypsum Plaster B 100.00 MM Air gap 10MM 13 MM Gypsum Plaster B **U-value: 1.946 W/m².K**		12.50 MM 2010 NCM Plasterboard 38.00 MM 2010 NCM insulation lambda 50.00 MM 2010 NCM Cavity Unventilated 12.50 MM plaster Board **U-value: 0.481 W/m².K**		12.50 MM 2010 NCM Plasterboard 38.00 MM 2010 NCM insulation lambda 50.00 MM 2010 NCM Cavity Unventilated 12.50 MM plaster Board **U-value: 0.481 W/m².K**
ROOF		13.00 MM Gypsum Plaster B 100.00 MM Air gap 10MM 13 MM Gypsum Plaster B **U-value: 2.13 W/m².K**		190.00 MM Asphalt 1 130.00 MM Fibreboard 44.00 MM XPS Extruded Polystyrene **U-value: 0.258 W/m².K**		190.00 MM Asphalt 1 130.00 MM Fibreboard 44.00 MM XPS Extruded Polystyrene **U-value: 0.258 W/m².K**

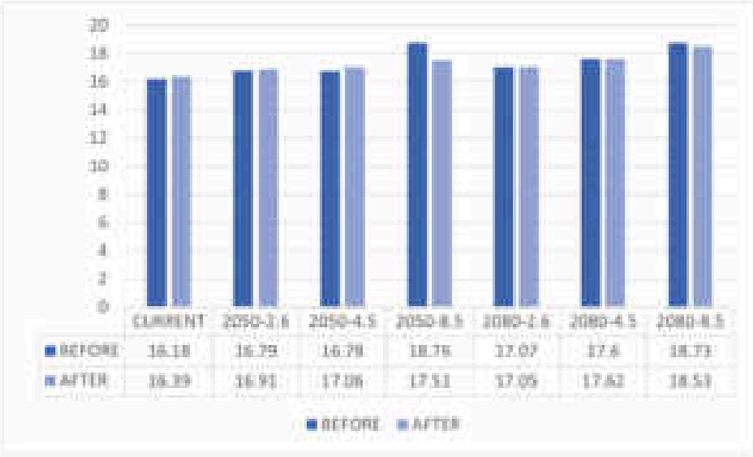
3.2.Window to Wall Ratio

GF

Highest temp

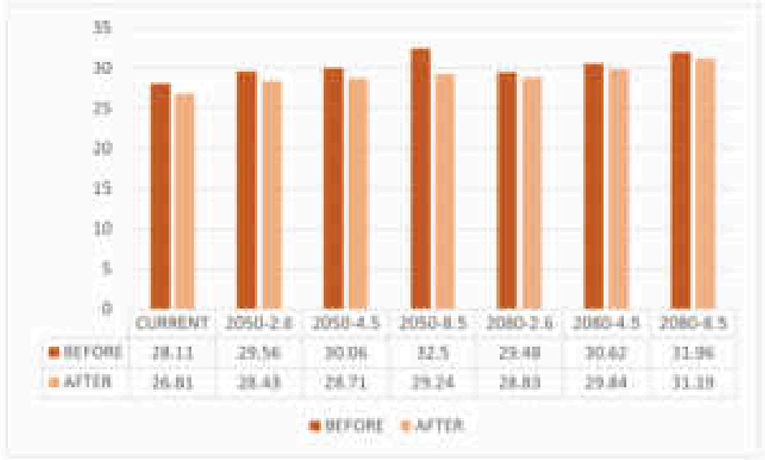


Lowest temp

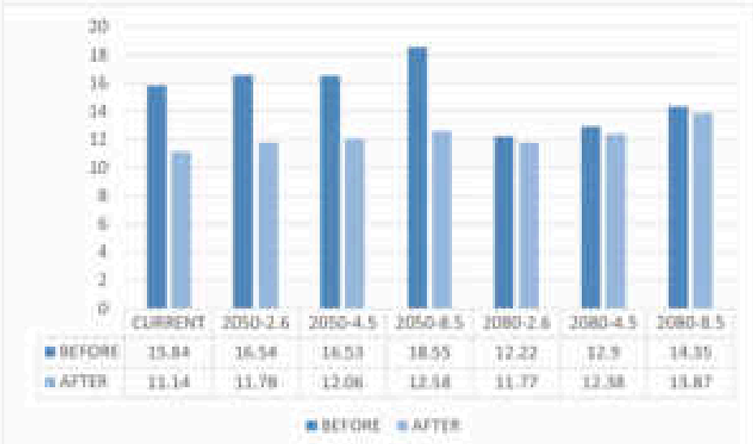


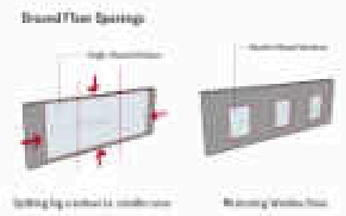
LF

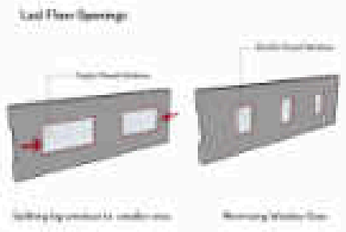
Highest temp



Lowest temp







Ground Floor:

Reduced window-to-wall ratio of the building shows a consistent reduction in operative temperature between 9.89%(2080 RCP 8.5) and 13.3%(2050 RCP 8.5)

It is observed that there is not much difference in the lowest temperature recordings. The operative temperatures increase slightly by 0.11% (2080 RCP 4.5) and 1.67% (2050 RCP 4.5) except for 2050 RCP 8.5, 2080 RCP 2.6 and 8.5 which records a reduction between 0.12%(2080 RCP 2.6) and 6.6%(2050 RCP 8.5).

Last Floor:

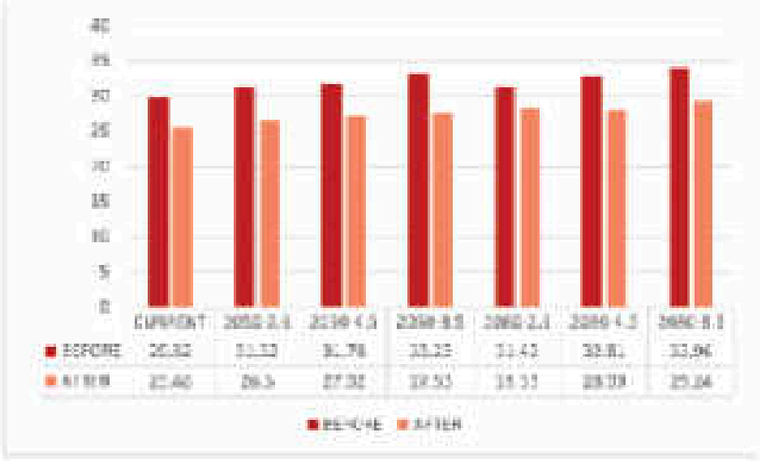
There is a record of moderate decrease in operative temperatures across all scenarios ranging from 2.2% to 10.03% (2050 RCP 8.5). This indicates that low window-to-wall ratio effectively mitigates overheating in future climates but the impact is less in extreme future climates of 2080 (2%-3%).

For the lowest temperature charts, there is significant reduction in operative temperatures between 27.05% and 32.2% (2050 RCP 8.5). The impact reduces in 2080 with lower percentages between 3.34% and 4.03%.

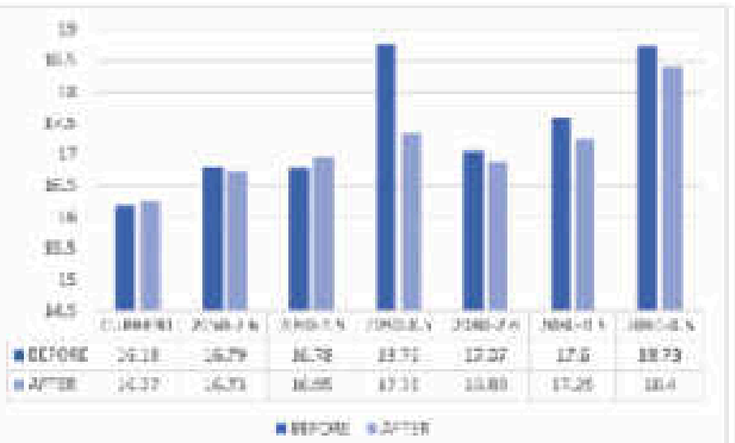
3.3. Horizontal Shading

GF

Highest temp

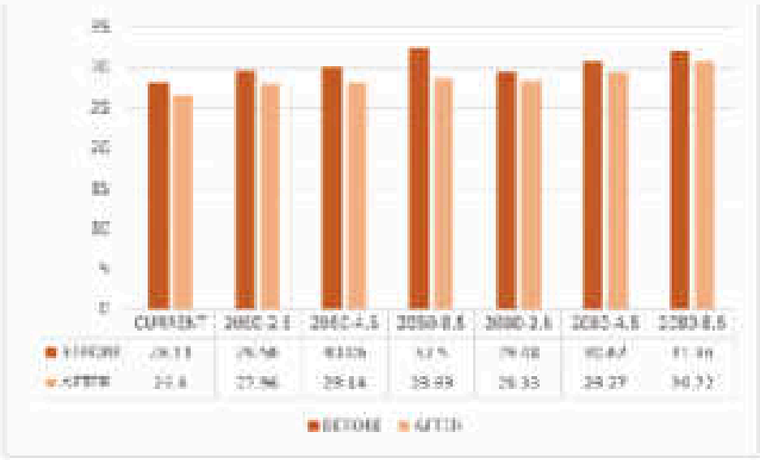


Lowest temp

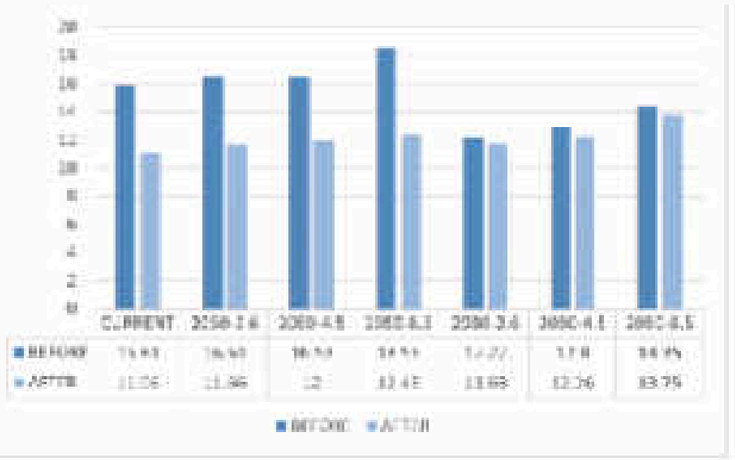


LF

Highest temp



Lowest temp





Ground Floor:

For highest temperature recording, the 1.5m shading effectively reduces the operative temperatures across the different climate conditions between 9.84%(2080 RCP 2.6) and 17.15%(2050 RCP 8.5).

At lowest temperatures, shading slightly reduces operative temperatures between 0.36%(2050 RCP 2.6) and 7.52%(2050 RCP 8.5). An exception is seen in current and 2050 RCP 4.5 which had slight increase of 0.56% and 1.01% respectively.



Last Floor:

Shading consistently reduces operative temperatures in all scenarios between 3.9% (2080 RCP 2.6) and 11.72% (2050 RCP 8.5)

Shading significantly impacts the low operative temperatures in all scenarios but majorly in current and 2050 climate with percentage reductions between 27.4%(2050 RCP 4.5) and 32.89% (2050 RCP 8.5). Less impact is recorded in 2080 climates with percentage decrease between 4.18%(2080 RCP 8.5) and 4.96%(2080 RCP 4.5).

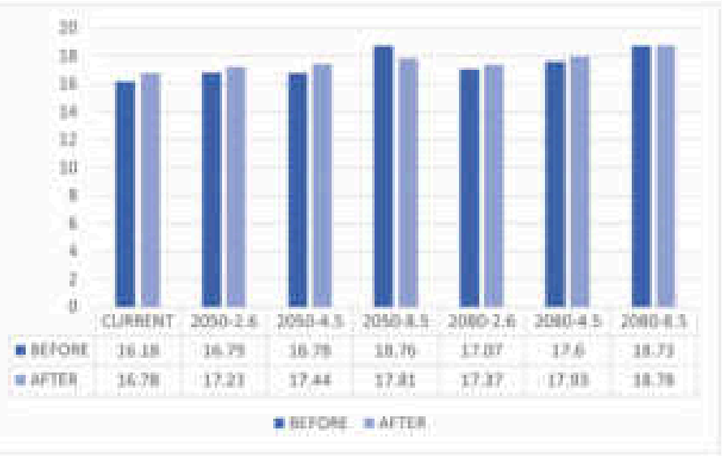
3.4. Window Glazing Upgrade

GF

Highest temp



Lowest temp



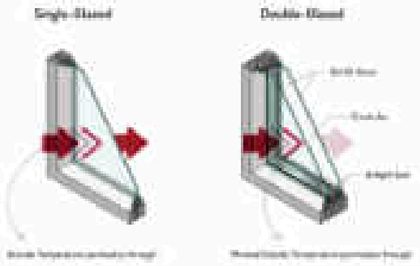
LF

Highest temp



Lowest temp





Ground Floor:

At the highest temperatures, incorporating a double-glazed clear glass window (6mm with 13mm air filled gap) contributes to significant drop in operative temperatures between 27.05% and 32.2%. The percentage decrease is lower in 2080 between 3.34% and 4.03%.

At lower temperatures, double glazing tends to slightly increase temperatures across all scenarios from 0.27% to 3.93% except for 2050 RCP 8.5 which has a 5.06% reduction in operative temperature.

Last Floor:

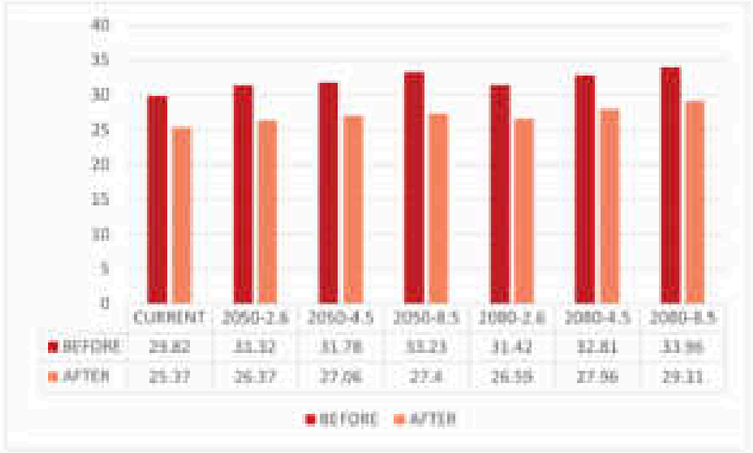
At high temperatures, double glazing is a good intervention which reduces operative temperatures with percentages between 4.16% (2080 RCP 8.5) and 11.54% (2050 RCP 8.5) across all scenarios.

Similarly, for lower temperatures, there is a significant reduction in temperatures between 29.9% (current) and 32.73% (2050 RCP 8.5). Slight reduction in operative temperatures occurs in 2080 with percentage decrease between 4.3% and 5.19%.

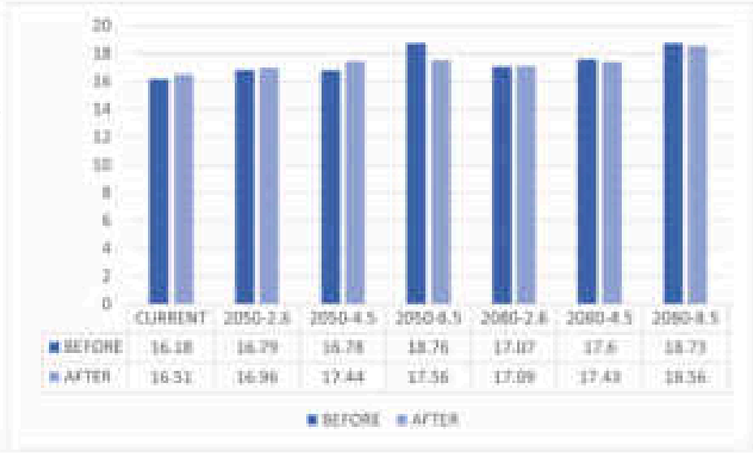
3.5. Night Ventilation

GF

Highest temp



Lowest temp

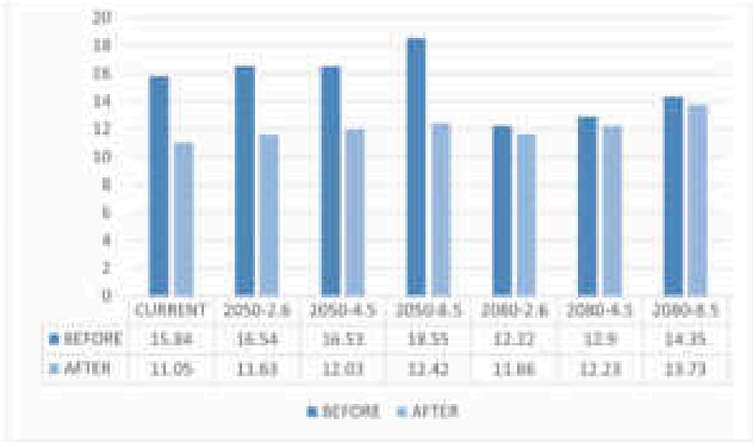


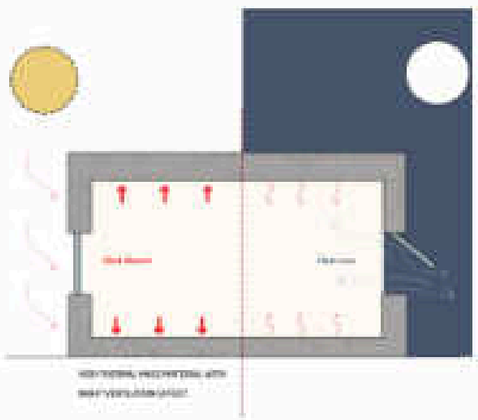
LF

Highest temp



Lowest temp





Ground Floor:

Night ventilation significantly reduces high temperatures in all climate scenarios with percentage reduction between 14.28% and 17.54%.

In lower temperature recordings, there is a slight increase in operative temperatures in four climate scenarios except 2050 RCP 8.5, 2080 RCP 4.5 and 2080 RCP 8.5 which has a 6.7%, 0.97% and 0.91% reduction in operative temperatures respectively.

Last Floor:

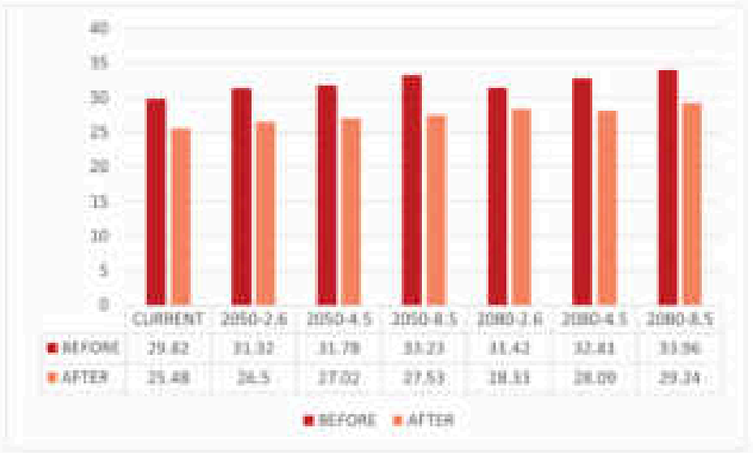
The impact of night ventilation across all scenarios leads to a consistent percentage reduction of temperatures between 4.04% (2080 RCP 8.5) and 11.91% (2050 RCP 8.5).

The lowest temperature recordings similarly shows a consistent reduction in temperatures across all scenarios with current and 2050 climates receiving significant percentage of temperature reduction between 27.22% and 33.05%. There is a percentage decrease for 2080 between 4.32% and 5.19%.

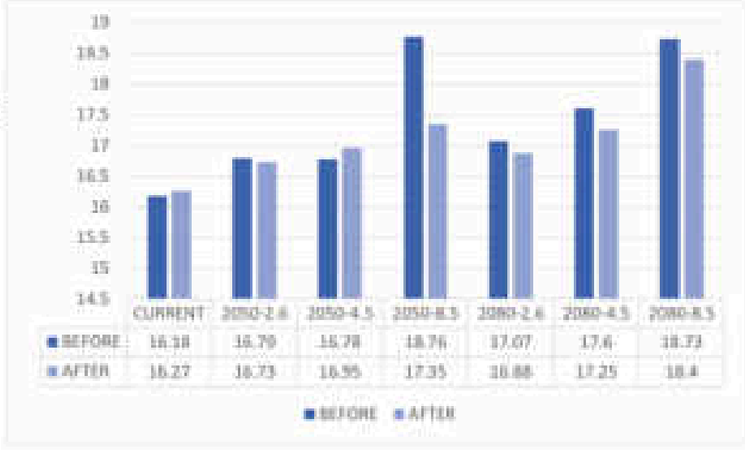
3.6. High-Thermal Mass Materials with Green Roof

GF

Highest temp



Lowest temp



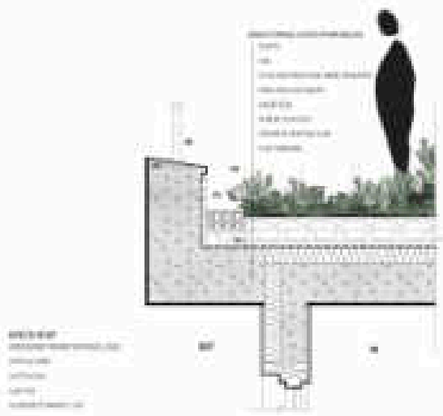
LF

Highest temp



Lowest temp





Ground Floor:
High thermal mass leads to a consistent reduction in high operative temperatures across all climate scenarios with percentage reduction between 13.63% (2080 RCP 8.5) and 16.91% (2050 RCP 8.5)
However, in lowest temperature recordings, thermal mass increases operative temperatures between 1.07% and 4.65% for all except 2050 RCP 8.5 which had a percentage reduction of operative temperature by 4.37%.

Last Floor:
Significant reduction of high operative temperatures is observed here in all scenarios between 5.47%(2050 RCP 8.5) and 12.98% (2050 RCP 8.5).
The recorded lowest operative temperatures slightly reduce for current and 2050 climate with a percentage reduction between 0.24% (2050 RCP 4.5) and 8.41%(2050 RCP 2.6). However, the same intervention shows a substantial increase in operative temperatures for 2080 climate with percentages between 26.62% (2080 RCP 8.5) and 31.86% (2080 RCP 3.5)

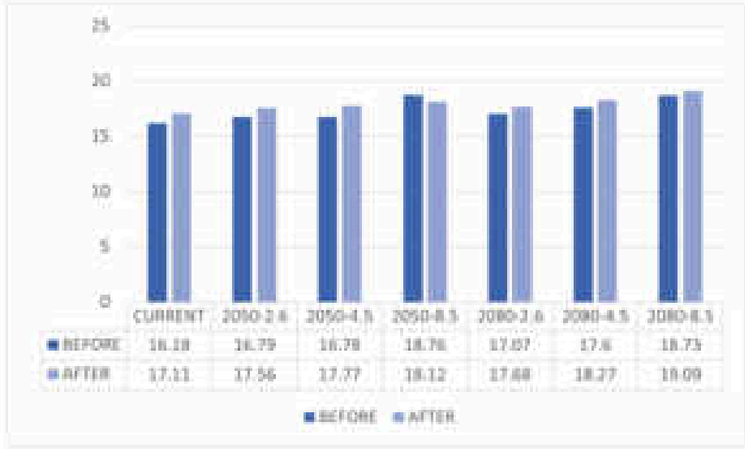
3.7. Low U-value Materials

GF

Highest temp



Lowest temp

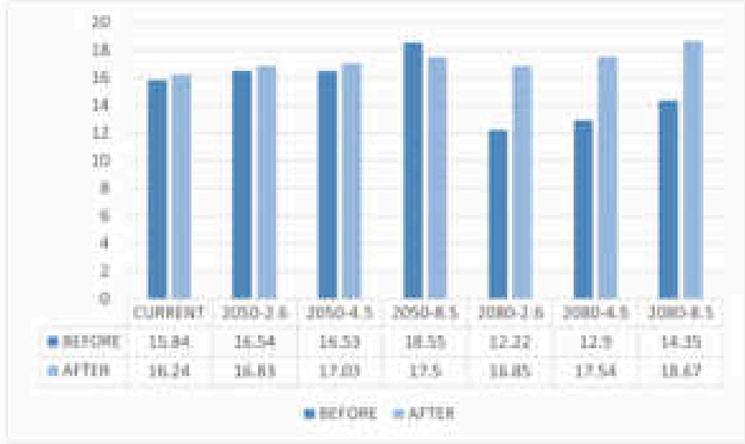


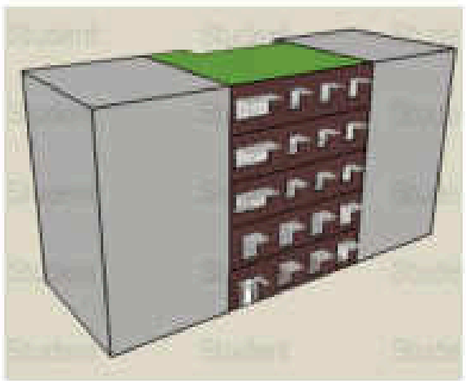
LF

Highest temp



Lowest temp





Ground Floor:

Choosing materials with low U-values for the envelope of the building significantly reduced the high operative temperatures for all scenarios with percentages between 13.87%(2080 RCP 8.5) and 17.03% (2050 RCP 8.5).

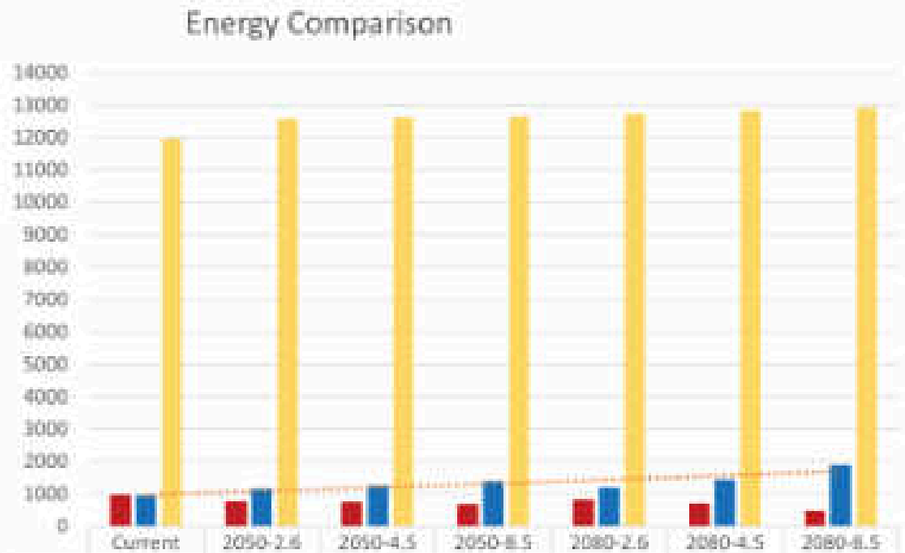
The recorded lower operative temperatures saw an increase in operative temperature for all climate scenarios with percentages between 1.92% (2080 RCP 8.5) and 5.90% (2050 RCP 8.5) except for 2050 RCP 8.5 which had a reduction in operative temperature by 3.41%.

Last Floor:

The high operative temperatures recorded has moderate temperature reduction between 5.13% (2080 RCP 4.5 & 8.5) and 12.62% (2050 RCP 8.5).

The lowest temperature recordings shows mild increase in operative temperatures between 1.75% and 3.03% for current, 2050 RCP 2.6 and 4.5. 2080 has a significant increase in operative temperatures between 30.1% and 37.89%. An exceptional case is seen for 2050 RCP 8.5 which records a reduction in operative temperature by 5.6%.

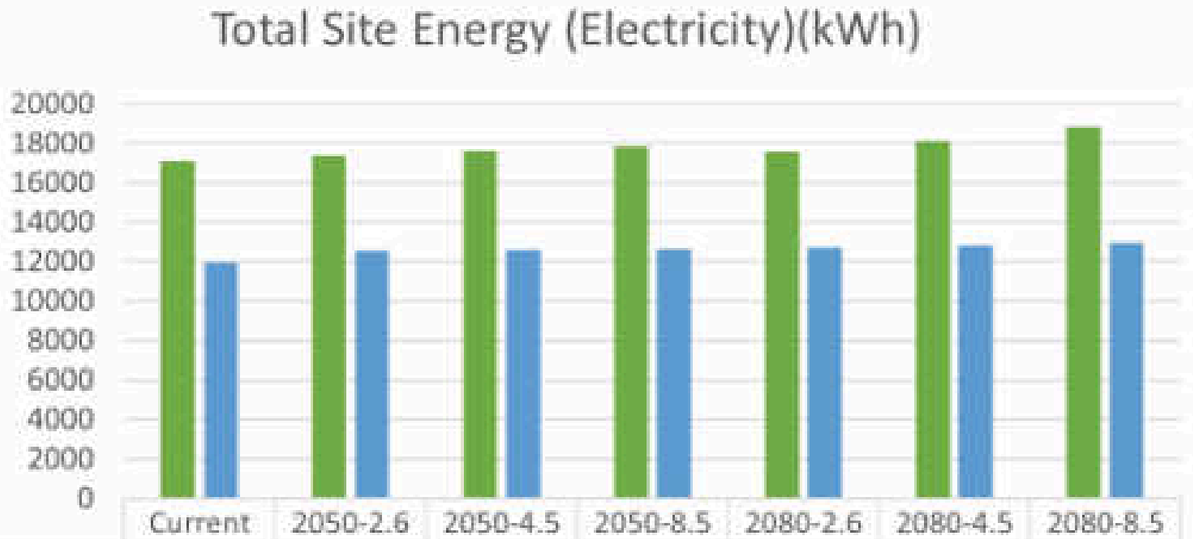
3.8. Energy



	Current	2050-2.6	2050-4.5	2050-8.5	2080-2.6	2080-4.5	2080-8.5
Heating (Electricity) (kWh)	950.8	770.91	753.33	674.85	822.5	711.23	485.49
Cooling (Electricity) (kWh)	931.22	1173.14	1269.64	1409.6	1201.88	1467.15	1902.01
Total Site Energy(Electricity) (kWh	11968.67	12570.14	12612.92	12656.26	12723.23	12839.15	12949.66

Energy Consumption:
The chart confirms the impact of rising temperatures across the climate scenarios. From the current weather to 2080 RCP 8.5, there is reduction in heating demand by 49% whiles cooling exponentially increases by 104% due to hotter summers.

Even though there is a drop in heating, overall site energy use steadily has a percentage increase between 5 to 8.2% across all scenarios.



	Current	2050-2.6	2050-4.5	2050-8.5	2080-2.6	2080-4.5	2080-8.5
BEFORE	17097.56	17373.53	17619.18	17842.8	17588.78	18106.44	18846.7
AFTER	11968.67	12570.14	12612.92	12656.26	12723.23	12839.15	12949.66

Total Site Energy:
The Heating, Ventilation and Air-conditioning system used across all scenarios is the Variable Refrigerant Flow (VRF) with Dedicated Outside Air System (DOAS). There is a significant reduction of energy use between the default model in each scenario, which excluded passive strategies and the final model with all passive strategies incorporated.

Even though VRF is an efficient mechanical system, the steady rise in temperature increases cooling demand during warmer months, especially in summer causing a percentage increase in energy use from 29.9% in the current temperature to 31.29% in 2080 RCP 8.5.

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